

Sales Forecasting & Demand Intelligence System

Executive Summary

This project analyzes the Superstore Sales dataset to forecast future sales, detect unusual sales behaviour, segment products based on demand, and classify customers using RFM analysis. Three forecasting models (SARIMA, Prophet, and XGBoost) were compared. Based on MAE, RMSE, and MAPE, SARIMA achieved the best performance and was selected for production forecasting.

Key Findings

- Technology generated the highest overall sales and is expected to continue leading future revenue.
- West and East regions contributed the strongest sales performance.
- Monthly sales showed an upward trend with seasonal fluctuations.
- Isolation Forest detected 11 anomalous weeks while Z-Score detected 2 extreme outliers.

Forecast Summary

The selected SARIMA model was used to generate a 36-month forecast for categories and regions. Technology is projected to remain the strongest category, while Furniture and Office Supplies show stable growth. Forecast confidence intervals indicate moderate uncertainty but an overall positive trend.

Product & Customer Segmentation

- Product clustering grouped products into High Demand, Medium Demand and Low Demand segments.
- Customer segmentation identified Premium, Regular and Low Value customers using RFM analysis.
- These segments can support inventory planning and targeted marketing campaigns.

Business Recommendations

1. Prioritize inventory for Technology products.
2. Retain premium customers with loyalty offers and personalized promotions.
3. Monitor detected anomalies to identify festive demand spikes or operational issues.

Limitations

Forecasts are based on historical sales patterns and may be affected by sudden market changes, promotions, economic events, or supply-chain disruptions. Periodic model retraining is recommended.